

Nervous Tissue

How one cell sends a message to another, and why it takes only milliseconds.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Name the parts of a neuron and say what each one does
- Explain why the inside of a resting neuron is negative
- List the steps of an action potential in order
- Say why myelin makes a signal faster
- Explain what crosses a synapse, and what happens on the other side

1 The cell that carries the message

A **neuron** has three working parts. Each part has one job.

- **Dendrites** are the branches. They *receive* messages from other cells.
- **The cell body** is the hub. It adds up what comes in and decides whether to pass it on.
- **The axon** is the long cable. It *carries* the message away to the next cell.

Neurons do not work alone. **Glial cells** outnumber them and do everything else. They feed neurons, clean up around them, and wrap the axons in insulation.

Keep that insulation in mind. It matters more than it sounds.

THINK OF IT THIS WAY

A neuron is shaped like a tree with one very long root. Messages land on the branches. The trunk decides if they add up to something worth sending. The root carries the answer a long way to someone else.

2 Why a resting neuron is negative inside

A neuron doing nothing is not really doing nothing. It is holding a charge, ready.

The cell spends energy pumping **sodium out** and **potassium in**. Potassium then leaks back out, and takes positive charge with it.

The result is that the inside ends up **more negative than the outside**. It sits at about **–70 millivolts**.

That difference is the **resting membrane potential**. It is stored energy, and it is why the cell can fire the instant it needs to.

THINK OF IT THIS WAY

It is a battery on a charger. Nothing is happening, but the charge is sitting there, ready to spend.

3 The action potential, in order

When a signal is strong enough, the neuron fires. Every time it fires, it does the same thing, the same size and the same shape.

- 1 **Something pushes the voltage up.** A message arrives, and the inside becomes a little less negative.
- 2 **Threshold is reached** at about –55 mV. This is the point of no return.
- 3 **Sodium rushes in.** Gates fly open and sodium floods the cell. The inside swings *positive*. This is **depolarization**.
- 4 **Potassium leaves.** Sodium gates shut, potassium gates open, and potassium exits. That drags the voltage back down. This is **repolarization**.
- 5 **A brief overshoot** takes the cell a bit more negative than resting. Then the pumps restore –70 mV, and it is ready to fire again.

All-or-none means exactly what it says. A stronger stimulus does not make a bigger action potential. It makes them fire *more often*.

That is how your nervous system codes "a little" versus "a lot." Not by size, but by how often.

4 Why myelin makes it faster

Drag the slider and watch what changes.

SIGNAL SPEED ALONG AN AXON



A bare axon has to rebuild the signal at *every single point* along its length. That is slow.

Now wrap it in **myelin**. The signal can only be rebuilt at the small gaps between the wrappings, called the **nodes of Ranvier**.

So it **jumps from gap to gap** instead of crawling. That jumping is called **saltatory conduction**, from the Latin word for leap.

THINK OF IT THIS WAY

Walking down a hallway and touching every tile, versus taking long strides and touching every tenth one. Same hallway, same destination, far less time.

This is also why damage to myelin matters. In multiple sclerosis the insulation is stripped away. The signal drops back to crawling, so messages arrive late or not at all.

5 Crossing the gap

Neurons never quite touch. There is always a gap, called the **synapse**.

An electrical signal cannot jump that gap. So the message changes form.

- 1 The action potential arrives at the end of the axon.
- 2 **Calcium enters**, and tiny sacs of chemical move to the membrane.

- 3 **Neurotransmitter is released** into the gap and drifts across.
- 4 It **binds to receptors** on the next cell, and opens channels there.
- 5 That either nudges the next cell *toward* firing, or *away* from it.

The next cell adds up everything landing on it. All the pushes toward firing, and all the pulls away.

If the total reaches threshold, it fires. If not, nothing happens. That adding up is called **summation**, and it is why a nervous system can make decisions instead of just passing noise along.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. A resting neuron sits at about -70 mV. What makes the inside negative?

- a) Sodium is pumped in and stays there
- b) Potassium leaks out and takes positive charge with it
- c) Chloride is pumped out of the cell
- d) The axon has no charge at rest

2. What happens first once threshold is reached?

- a) Potassium leaves the cell
- b) Sodium rushes into the cell
- c) The pump shuts down
- d) Neurotransmitter is released

3. A stimulus gets much stronger. What does the neuron do?

- a) Fires a bigger action potential
- b) Fires action potentials more often
- c) Fires more slowly
- d) Stops firing to protect itself

4. Why does a myelinated axon carry a signal faster than a bare one?

- a) Myelin adds extra sodium channels along the whole axon
- b) The signal jumps between gaps instead of being rebuilt everywhere
- c) Myelin makes the axon shorter
- d) Myelin pushes the signal along like a pump

5. What actually crosses the synapse from one neuron to the next?

- a) The action potential itself
- b) An electrical spark
- c) A chemical neurotransmitter
- d) Calcium from the next cell

Answer key

1. **Potassium leaks out and takes positive charge with it**

Right. The pump sets the gradients, and potassium leaking out leaves the inside more negative.

2. **Sodium rushes into the cell**

Correct. Sodium gates fly open and the inside swings positive.

3. **Fires action potentials more often**

Right. Strength is coded by frequency, not by size.

4. **The signal jumps between gaps instead of being rebuilt everywhere**

Correct. That jumping between nodes is saltatory conduction.

5. **A chemical neurotransmitter**

Right. Neurotransmitter drifts across and binds receptors on the next cell.

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